TRANSACTION MONITORING ALERT CASE SUMMARY

A practical introduction to machine learning, alerts, cases and cross-border transaction analysis

1. What is Machine Learning (ML)?

Machine Learning is a branch of artificial intelligence in which a system learns patterns from data to make predictions, rank risks or identify unusual behaviour. Instead of relying only on manually written rules, an ML model can learn what typical activity looks like from historical examples and flag transactions that differ from expected patterns.

ML concept	Meaning in transaction monitoring
Features	Inputs such as amount, currency, country, customer type, transaction frequency and prior behaviour.
Model / score	A model estimates risk or unusualness; a high score is a signal to review, not proof of wrongdoing.
Alert	A rule or model output that highlights activity for review by an analyst.
Human investigation	An investigator checks context, documents, counterparties and explanations before deciding the outcome.

Important: ML does not decide that a customer committed a crime. It supports prioritisation. Analysts must validate the evidence, consider legitimate explanations and follow internal policy.

2. Alert vs. case — what is the difference?

Alert: a signal raised by a monitoring rule or model because a transaction or pattern meets risk criteria. **Case:** an investigation record that groups one or more related alerts, transactions, customers and supporting evidence so an analyst can assess the activity as a whole.

3. Case scope and portfolio snapshot

This report uses a fictional three-month portfolio (1 January–31 March 2024) and one illustrative cross-border case. All names, amounts and metrics are synthetic and provided only for learning how to summarise an alert investigation.

Metric	Illustrative value	What it means
Alerts generated	1,245	Signals raised by monitoring controls
Cases created	320	Investigation records opened
Cases closed	305	Cases with a recorded disposition
Suspicious Activity Reports (SARs)	28	Cases escalated for suspicious activity reporting
No Further Action (NFA)	277	Cases closed without further escalation in this example
Average investigation time	4.2 days	Illustrative average from opening to closure
Transactions in example case	18	Transactions linked to the selected case review
Customers / entities linked	3	One customer and two related counterparties/entities

How to read the numbers

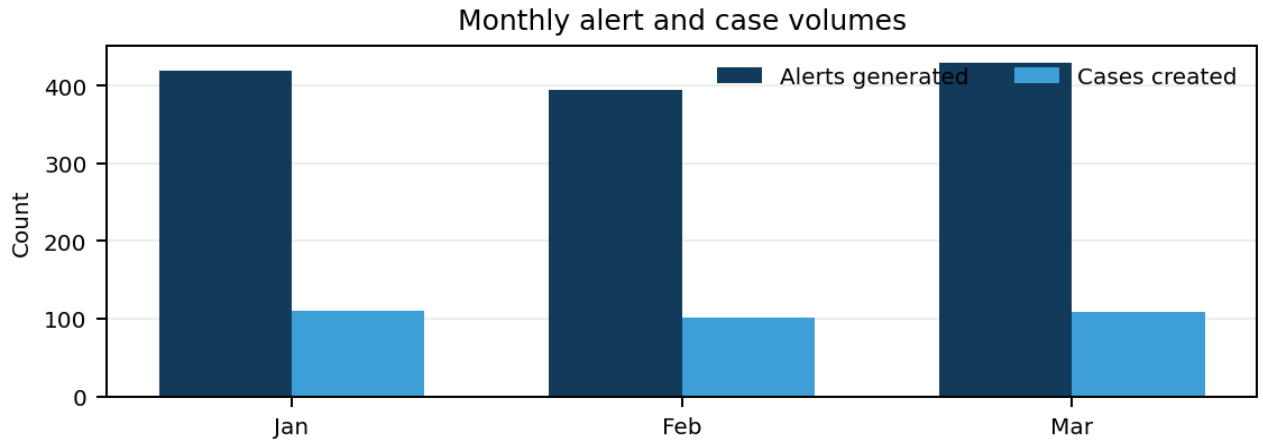
Alerts and cases are not interchangeable: multiple alerts can be grouped into one case, and not every alert becomes a case. SAR and NFA are example dispositions and must be applied according to jurisdictional requirements and internal procedures. The sample totals are illustrative, not operational benchmarks.

Case definition checklist

Case element	What should be recorded
Identity	Case ID, customer ID, customer type, industry, account / entity references
Activity	Transaction count, total value, currencies, dates, originator and beneficiary countries
Risk indicators	Alert typology, model/rule reason, unusual behaviour, relevant thresholds
Evidence and decision	Documents checked, explanation sought, findings, reviewer and final disposition

4. Alert and case trends

The chart compares alerts generated with cases created across the illustrative reporting period. The gap is expected because not every alert is converted into a case; triage rules, duplicate alerts, thresholds and review procedures influence that conversion.



Illustrative closed-case outcome	Count	Share of closed cases
No Further Action (NFA)	277	90.8%
SAR filed / escalated	28	9.2%
Total closed cases	**305**	**100%**

Analyst interpretation

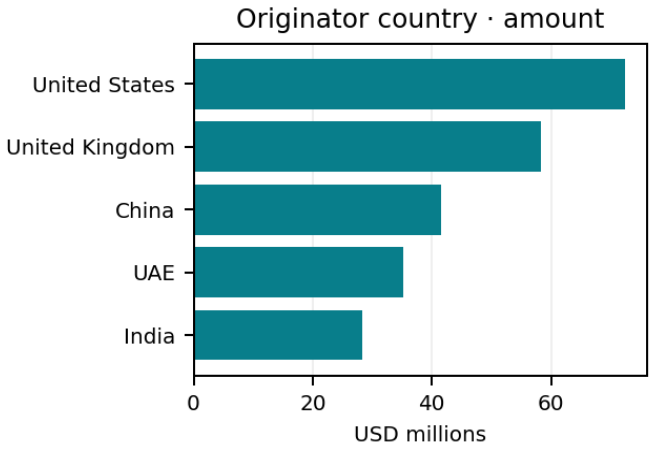
A rising alert count does not automatically mean financial crime has increased. Changes can reflect new scenarios, threshold tuning, seasonality, data quality or customer activity. Monitor alert-to-case conversion, investigation time, outcome distribution, repeat alerts and quality-review findings together.

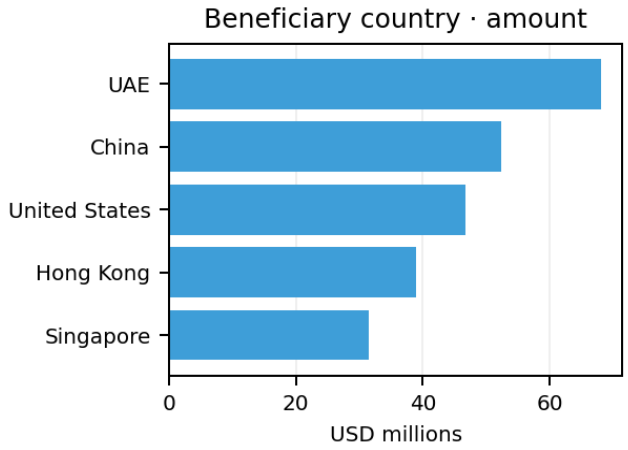
Illustrative alert themes

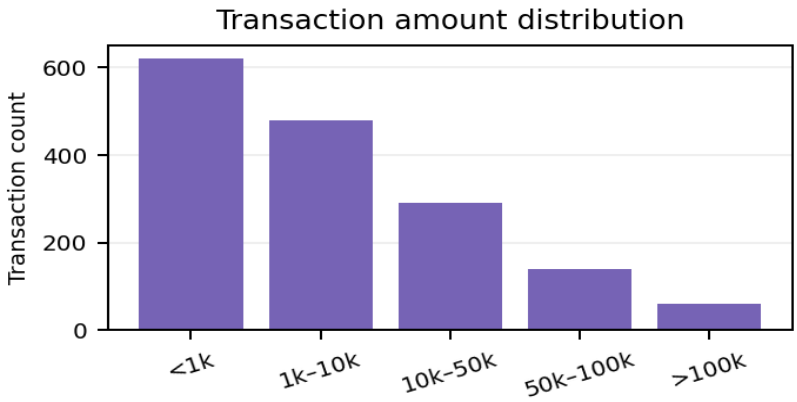
Alert theme	Share of alerts	Possible investigation question
High-risk country exposure	28%	Is the country exposure justified by the customer’s business and counterparties?
Large amount	24%	Is the value consistent with expected turnover and historical behaviour?
Potential structuring	18%	Are multiple transactions arranged to avoid controls or thresholds?
Unusual pattern	15%	What changed from the customer’s normal activity?
Rapid movement / other	15%	Do funds move quickly through related parties without a clear rationale?

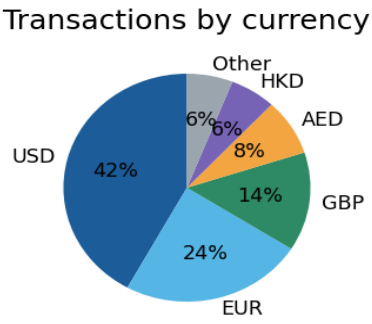
5. Transaction analysis: countries, amounts and currencies

Country exposure should be assessed using the full payment chain, not only the customer's location. Originator and beneficiary country fields may reflect different parties in the payment message; intermediary banks and related entities can add further context.









Measure	Illustrative value	How to use it
Transactions reviewed	18	Check whether all transactions are in scope and deduplicated.
Gross transaction value	USD 1.59m equivalent	Confirm FX rates, value dates and whether reversals are included.
Largest transaction	USD 250,000	Compare with expected activity and documented business purpose.
Countries shown	US, UK, China, UAE, India, Hong Kong, Singapore	Validate country fields and identify intermediaries / beneficial parties.

Interpretation guardrail: a country or a large amount alone is not evidence of illicit activity. Consider customer profile, transaction purpose, counterparties, expected behaviour, source of funds, supporting documentation and applicable risk methodology.

6. Example alert case: investigation summary

Case ID: C-2024-0158 | Alert types: High-risk country exposure + large amount | Status: Escalated for suspicious activity reporting (illustrative)

Field	Case record	Field	Case record
Customer ID	CUST-987654	**Customer type**	Corporate
Industry	Trading	**Case opened**	15 Feb 2024
Case closed	22 Feb 2024	**Investigation time**	5 days (working-time convention may vary)
Transactions reviewed	18	**Related entities**	2 counterparties / entities
Total reviewed value	USD 1.59m equivalent	**Disposition**	SAR filed / escalated (example)

Illustrative transaction network



Date	Originator country	Beneficiary country	Amount (USD)	Currency
12 Feb 2024	United States	United Arab Emirates	250,000	USD
13 Feb 2024	United Arab Emirates	Hong Kong	180,000	USD
14 Feb 2024	United States	Hong Kong	95,000	USD

Investigation summary

The illustrative corporate customer initiated multiple high-value cross-border transfers involving the United States, the United Arab Emirates and Hong Kong over a short period. The pattern was flagged because the value and jurisdictional routing required review against the customer's stated trading activity and expected payment profile. The investigator should validate invoices and contracts, confirm the economic purpose and counterparties, assess source of funds, review related accounts and intermediaries, and document any inconsistencies. In this synthetic example, unresolved concerns led to escalation for suspicious activity reporting.

Key conclusion: an alert is a starting point, not a finding. A defensible case summary connects the trigger, transaction facts, customer context, evidence reviewed, alternative explanations and reasoned disposition. Never infer criminality from geography or transaction value alone.

Training note: all IDs, dates, values, volumes and outcomes in this report are fictional and internally illustrative. They must not be used as real financial-crime statistics, policy guidance or a live investigation record.